
TECHNICAL SPECIFICATIONS
24", 18", and 12" Butterfly Valves with Electric Actuators and Flange Kits

1.0 INTENT

The following specifications pertain to the purchase of one (1) 24 inch butterfly valve with an electric actuator, one (1) 18 inch butterfly valve with an electric actuator, and one (1) 12 inch butterfly valve with an electric actuator and corresponding flange kits for the three valves for Little Goose Lock and Dam.

2.0 SCOPE

The successful offeror must provide one (1) new 24 inch flanged butterfly valve with one (1) electric actuator and one (1) 24 inch flange kit; one (1) new 18 inch wafer style butterfly valve with one (1) electric actuator and one (1) 18 inch flange kit; one (1) new 12 inch wafer style butterfly valve with one (1) electric actuator and one (1) 12 inch flange kit. All valves, actuators and flange kits must conform to the specifications listed in section 3. Technical documentation, including part numbers and a configuration drawing of the complete assembly must be submitted to the technical point of contact at site (through the Contracting Office), for technical evaluation **prior to contract award**.

3.0 PRODUCT REQUIREMENTS

ITEM	DESCRIPTION	QTY	U/M
1	24", 150B pressure class, Flanged Butterfly valve, DI body, DI disk, 316ss shaft	1	EA
2	Electric actuator for the 24" valve, 208, 3ph, 4-20ma, includes remote controls mounting kit with 40ft controls cable	1	EA
3	24" 150# flange kit - qty (20) zinc plated 1-1/4"-7 x 6" ASTM A307 grade A class 2A bolts with corresponding class 2B nuts	1	EA
4	18", 150B pressure class, Wafer Butterfly Valve, DI body, DI disk, 316ss shaft	1	EA
5	Electric actuator for the 18" valve, 208, 3ph, 4-20ma, includes remote controls mounting kit with 40ft controls cable	1	EA
6	18" 150# flange kit - qty (16) zinc plated 1-1/8"-7 x 8" ASTM A307 grade A class 2A bolts with corresponding class 2B nuts	1	EA
7	12", 150B pressure class, Wafer Butterfly Valve, DI body, DI disk, 316ss shaft	1	EA
8	Electric actuator for the 12" valve, 208, 3ph, 4-20ma, includes remote controls mounting kit with 40ft controls cable	1	EA
9	12" 150# flange kit - qty (12) zinc plated 7/8-9 x 7" ASTM A307 grade A class 2A bolts with corresponding class 2B nuts	1	EA

3.1 PRODUCT SPECIFICATIONS

The new 24 inch flanged butterfly valve, suitable for river water with a temperature range from 33°F to 85°F must have a body constructed of ASTM A536, Class B ductile iron and meet the 150B pressure class. The flanges must be fully faced and drilled in accordance with ANSI B16.1 class 125 flange and bolting connections. The valve must have an EPDM rubber seat and suitable for bi-directional shutoff at rated pressure. The valve seat must be retained in the valve body by mechanical means without retaining rings, segments, screws or hardware of any kind in the flow stream. The valve disc must be

constructed of ASTM A536 class B ductile iron with a stainless steel seating edge. The disc must not have any hollow chambers that can entrap water. The disc and shaft connection must be made with stainless steel pins. The valve shaft must be constructed of ASTM A-276 type 316 stainless steel. The flange face to flange face dimension must be 8 inches. The coating on the valve must be epoxy with a minimum of 8 mils thickness.

The electric actuator must mount to the 24 inch butterfly valve listed above and provide the required torque to operate the valve under full flow conditions. The actuator must be quarter turn with a modulating duty operating mode. The input voltage must be 208, 3-phase, 60 Hz with 4-20 mA analog output. The actuator and controls at a minimum must be IP68 rated and be able to withstand operating conditions between -20 °F and +150°F. The time to cycle the valve from fully closed to fully opened must be no less than 45 seconds and no more than 80 seconds. The controls must be capable of remote mounting from the actuator and include additional components and mounts including 40 feet of controls cable to allow for the remote controls mounting to a wall. The actuator must be equipped with a handwheel to provide a manual override option when power is not supplied. When installed, the handwheel must face perpendicular to the flow of water through the pipe in the X, -X, or Z direction per document 20250206. The dimension from the center of the valve stem to the furthest point on the actuator in the X, -X, or Y direction must be less than 27 inches. See document 20250206 for reference.

The flange kits must be for a 24 inch ANSI B16.1 class 125 flange. Each kit will include a full-face Aramid/Buna-N flange gasket that is 1/8" thick and qty (20) 1-1/4"-7 x 6" ASTM A307 grade A class 2A bolts with corresponding class 2B nuts.

The new 18 inch wafer style butterfly valve, suitable for river water with a temperature range from 33°F to 85°F must have a body constructed of ASTM A536, Class B ductile iron and meet the 150B pressure class. The Rubber valve seat must be of one-piece construction, molded and bonded into the valve body. Seats may not be located on the disc or be retained by segments and/or screws. The seat must cover the entire inner surface of the valve body and extend over the outside face of the valve body to form a flange gasket. The valve disc must be constructed of ASTM A536 class B ductile iron with a stainless steel seating edge. The disc must not have any hollow chambers that can entrap water. The disc and shaft connection must be made with stainless steel pins. The valve shaft must be constructed of ASTM A-276 type 316 stainless steel. The flange face to flange face dimension must be 4-11/16 inches. The coating on the valve must be epoxy with a minimum of 8 mils thickness.

The electric actuator must mount to the 18 inch butterfly valve listed above and provide the required torque to operate the valve under full flow conditions. The actuator must be quarter turn with a modulating duty operating mode. The input voltage must be 208, 3-phase, 60 Hz with 4-20 mA analog output. The actuator and controls at a minimum must be IP68 rated and be able to withstand operating conditions between -20 °F and +150°F. The time to cycle the valve from fully closed to fully opened must be no less than 45 seconds and no more than 80 seconds. The controls must be capable of remote mounting from the actuator and include additional components and mounts including 40 feet of controls cable to allow for the remote controls mounting to a wall. The actuator must be equipped with a handwheel to provide a manual override option when power is not supplied. When installed, the handwheel must face perpendicular to the flow of water through the pipe in the X, or Z direction per document 20250206. The dimension from the center of the valve stem to the furthest point on the actuator in the X, or Y direction must be less than 27 inches. See document 20250206 for reference.

The flange kits must be for an 18 inch ANSI B16.1 class 125 flange. The kit will include qty (16) 1-1/8"-7 x 8" ASTM A307 grade A class 2A bolts with corresponding class 2B nuts.

The new 12 inch wafer style butterfly valve, suitable for river water with a temperature range from 33°F to 85°F must have a body constructed of ASTM A536, Class B ductile iron and meet the 150B pressure class. The Rubber valve seat must be of one-piece construction, molded and bonded into in the valve body. Seats may not be located on the disc or be retained by segments and/or screws. The seat must cover the entire inner surface of the valve body and extend over the outside face of the valve body to

form a flange gasket. The valve disc must be constructed of ASTM A536 class B ductile iron with a stainless steel seating edge. The disc must not have any hollow chambers that can entrap water. The disc and shaft connection must be made with stainless steel pins. The valve shaft must be constructed of ASTM A-276 type 316 stainless steel. The flange face to flange face dimension must be 3-7/16 inches. The coating on the valve must be epoxy with a minimum of 8 mils thickness.

The electric actuator must mount to the 12 inch butterfly valve listed above and provide the required torque to operate the valve under full flow conditions. The actuator must be quarter turn with a modulating duty operating mode. The input voltage must be 208, 3-phase, 60 Hz with 4-20 mA analog output. The actuator and controls at a minimum must be IP68 rated and be able to withstand operating conditions between -20 °F and +150°F. The time to cycle the valve from fully closed to fully opened must be no less than 45 seconds and no more than 80 seconds. The controls must be capable of remote mounting from the actuator and include additional components and mounts including 40 feet of controls cable to allow for the remote controls mounting. The actuator must be equipped with a handwheel to provide a manual override option when power is not supplied. When installed, the handwheel must face perpendicular to the flow of water through the pipe in the X or Z direction per document 20250206. The dimension from the center of the valve stem to the furthest point on the actuator in the X, or Y direction must be less than 27 inches. See document 20250206 for reference.

The 12 inch flange kit must be for a 12 inch ANSI B16.1 class 125 flange. The kit will include qty (12) zinc plated 7/8-9 x 7" ASTM A307 grade A class 2A bolts with corresponding class 2B nuts.

4.0 WARRANTY

The valves must be covered by a minimum of one-year warranty on all parts, labor, and workmanship from factory defects.

The actuators must be covered by a minimum of one-year warranty on all parts, labor, and workmanship from factory defects.

5.0 INSTALLATION

The factory representative and/or a contractor's service technician must furnish instructions for recommended installation of the valve and actuator and include instructions for the recommended commissioning process.

6.0 OPERATION AND MAINTENANCE MANUALS

Two complete physical copies and a single digital copy of the Operation and Maintenance Manual for the valve and actuator will be required. The manual must include instructions on the installation, operation, maintenance, repair, troubleshooting, and warranty of the systems. They must also include diagrams listing all system components, part numbers, and a list of recommended spare parts.

Also included must be the name, phone, and email address of the contractor, manufacturer, and a distributor where spare parts can be purchased.

The following identification must be inscribed on the covers:

OPERATING AND MAINTENANCE INSTRUCTIONS

Contract Name

Little Goose Lock and Dam

Name of the Contractor

Contract Number

7.0 SHIPMENT AND DELIVERY

All items must be delivered FOB Destination to Little Goose Dam within 34 weeks of award. The contractor must be responsible for packing and shipping the components in a manner that will ensure undamaged and complete arrival to Little Goose Dam. No shipments are allowed other than normal business hours of Monday-Thursday 7:00 AM-3:00PM, excluding holidays.

All shipments and invoices must be delivered to:

Little Goose Dam

[REDACTED]

1001 Little Goose Dam Road

Dayton, WA 99328

[REDACTED]
